

Supplementary material

Deciphering the multivariable interactions of MOFs in heavy metal adsorption
via explainable machine learning for sustainable and resource-efficient
adsorbent development

Supplementary material includes:

22 pages, 4 texts, 5 tables, 5 figures

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Text S1. Details of data preprocessing

To ensure data quality and consistency across heterogeneous literature sources, a structured preprocessing pipeline combining rule-based filtering and domain-informed screening was employed. The objective was to balance data completeness, representativeness, and statistical reliability, while minimizing the introduction of bias due to excessive imputation (Fig. T1).

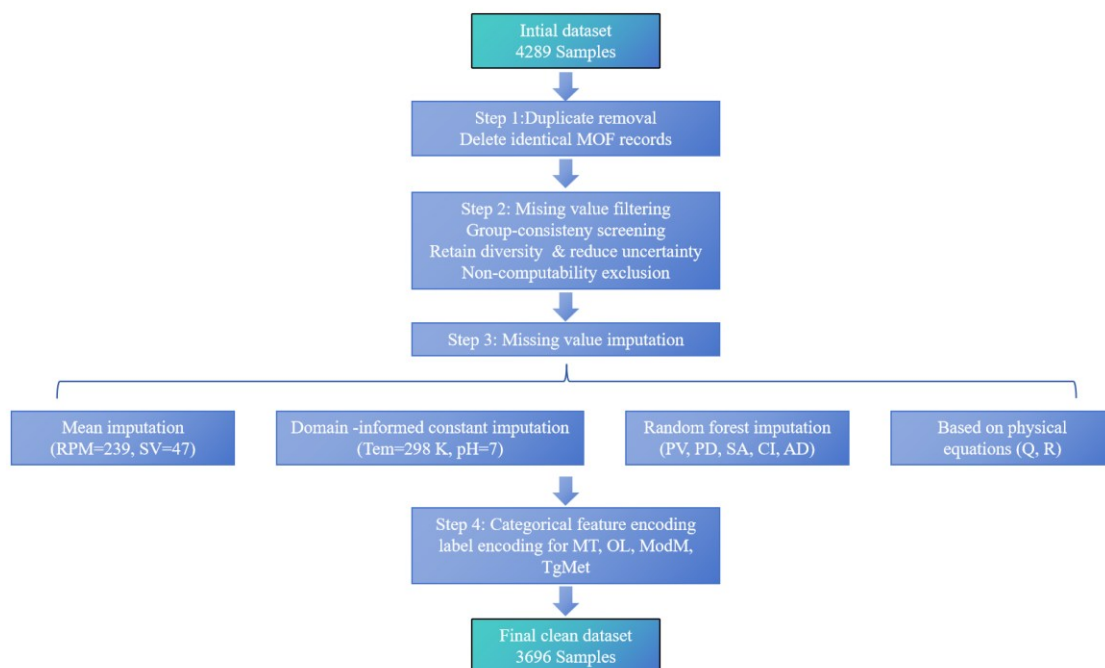


Fig. T1. Data preprocessing flowchart

The initial dataset comprised 4,289 samples collected from approximately 100 peer-reviewed studies. Data reduction to 3,696 samples was achieved through a multi-step filtering process:

(1) Duplicate removal: Identical records (same MOF composition, experimental conditions, and target outputs) were removed.

(2) Missing-value-based filtering: Samples with excessive missing information in key MOF structural descriptors (e.g., SA, PD, PV) were selectively excluded. Instead of applying a rigid threshold, a group-consistency strategy was adopted. If there are missing values in Q and R and the core parameters used to calculate Q and R are missing, the sample is directly excluded:

1. For experimental groups sharing identical MOF composition and similar conditions, if structural descriptors were entirely missing, a representative subset of samples was retained to preserve group-level information.

2. When only a small number of descriptors were missing, samples were largely retained to maintain statistical diversity.

3. As the degree of missingness increased, partial down-sampling within the group was applied to reduce uncertainty propagation.

This strategy reflects a trade-off between data completeness and information redundancy,

ensuring that the retained dataset preserves both diversity and reliability.

(3) Missing value imputation:

1. For the missing values of RPM and SV, mean imputation was performed using 239 rpm and 47 mL respectively, this approach maintains the overall sample mean stability while avoiding introducing additional variability, as these parameters are typically treated as independent experimental conditions rather than systematically investigated variables in different studies.

2. For variables with moderate missing ratios and potential nonlinear relationships (e.g., PV, PD, SA, CI, AD), a Random Forest-based imputation method was employed, leveraging multivariate correlations among features.

3. For parameters with commonly accepted standard conditions in adsorption studies (e.g., $\text{pH} \approx 7$, $T_{\text{em}} \approx 298 \text{ K}$), domain-informed constant imputation was adopted. First, from the perspective of data distribution, the T_{em} variable exhibits a highly concentrated distribution in the dataset, with both the upper and lower quartiles at 298 K (see Table S3 and Fig. S1), indicating that the vast majority of experiments were conducted under room temperature conditions. Moreover, this value is the most commonly adopted operating temperature in adsorption experiments, and studies without explicit temperature control often default to this setting. Therefore, uniformly setting missing values to 298 K effectively reflects the mainstream experimental conditions of the data without introducing additional bias.

Second, $\text{pH}=7$, as a neutral condition, is one of the most common baseline settings in water treatment experiments to avoid the destruction of MOF structures or excessive interference with adsorption mechanisms caused by extreme acidic or alkaline environments. Particularly in studies where pH is not explicitly reported, the conditions are typically assumed to be neutral or near-neutral. Thus, this setting has a reasonable empirical basis in the field of environmental engineering.

4. Derivation of adsorption capacity and removal rate.

The relationship between adsorption capacity (Q , mg g^{-1}) and removal rate (R , %) was established based on standard adsorption mass balance equations.

The adsorption capacity is defined as:

$$Q = \frac{(C_0 - C_e) V}{m}$$

where $C_0 (\text{mg L}^{-1})$ and $C_e (\text{mg L}^{-1})$ are the initial and equilibrium concentrations of the pollutant, respectively, $V (\text{L})$ is the solution volume, and $m (\text{g})$ is the mass of the adsorbent.

The removal rate is defined as:

$$R = \frac{C_0 - C_e}{C_0} \times 100\%$$

By combining the above equations, the adsorption capacity can be expressed as a function of removal rate:

$$Q = \frac{R}{100} \cdot \frac{C_0 \cdot V}{m}$$

In this study, the adsorbent dosage is defined as:

$$AD = \frac{m}{V}$$

where AD is expressed in $mg\ L^{-1}$. Since adsorption capacity Q is defined in $mg\ g^{-1}$, while m in AD is expressed in mg , a unit conversion factor of 1000 is required to ensure dimensional consistency. Substituting $AD = m/V$ and applying the unit conversion yields:

$$Q = \frac{R}{100} \cdot \frac{C_0 \cdot 1000}{AD}$$

and equivalently:

$$R = \frac{Q \cdot AD}{1000 \cdot C_0} \times 100$$

These equations were used to derive missing values of Q or R when one of the two quantities was reported in the original literature and sufficient experimental parameters were available.

Samples lacking sufficient information for this calculation were excluded from the dataset to ensure data reliability. Additionally, the calculated values were cross-checked against reported values (when available) to ensure consistency.

This hybrid imputation strategy integrates statistical robustness with domain knowledge, reducing bias compared to single-method imputation.

(4) Categorical feature encoding: Categorical variables, including coordination metal (MT), organic ligand (OL), modification method (ModM), and target metal (TgMet), were encoded using label encoding to ensure compatibility with tree-based models. Given that XGBoost is insensitive to monotonic transformations of categorical indices, label encoding does not introduce artificial ordinal relationships in this context.

Overall, the preprocessing pipeline was designed to prioritize data reliability and structural consistency while preserving sufficient sample diversity for robust machine learning modeling. The final dataset (3,696 samples) reflects a balance between data quality control and information retention.

Text S2. Robustness and stability analysis of model performance and interpretability

We conducted a comprehensive robustness analysis, including repeated random splits of the training/test sets, subset evaluations of different target heavy metals in the final test set, and stability analysis of feature importance (SHAP interpretation stability). The results consistently demonstrated that predictive performance and interpretability findings remained stable across different data partitioning strategies (Fig. T2).

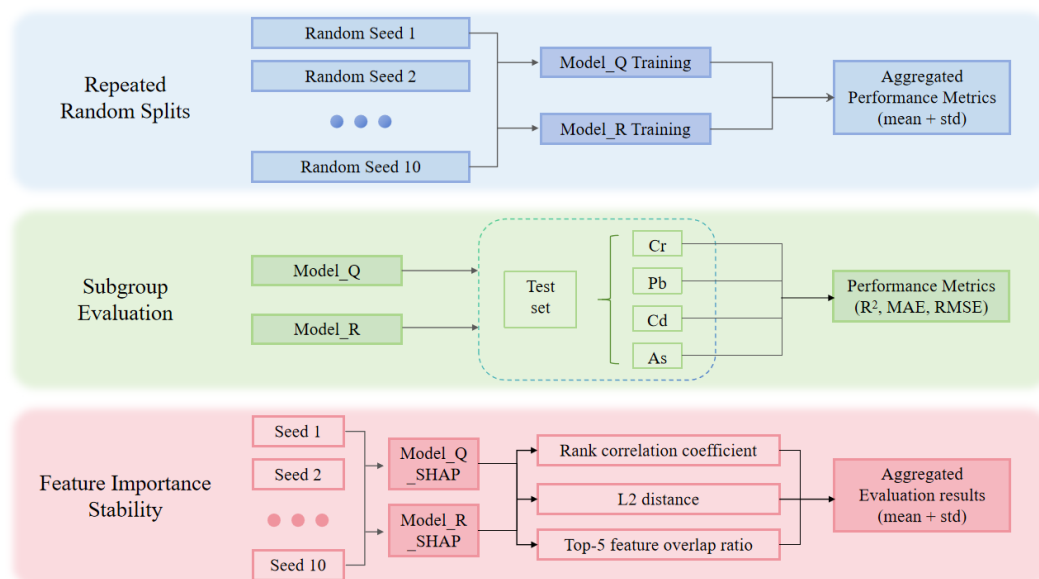


Fig. T2. Schematic diagram of robustness and stability analysis

1. Robustness analysis via repeated random splits. To evaluate the stability of the model with respect to data partitioning, we performed repeated random train/test splits using 10 different random seeds, with a fixed ratio of 90% for training and 10% for testing. For each split, the model was trained and evaluated under the same configuration, and the predictive performance was recorded. The results are summarized in terms of mean $\pm$ standard deviation. Specifically, the model achieves an R^2 of 0.92 ± 0.034 for Q and 0.901 ± 0.021 for R across different random splits. A boxplot of the R^2 values is also provided to visualize the distribution of performance (Fig. T3). These results indicate that the model performance is stable under different data partitions and not dependent on a particular random split, supporting the robustness and reliability of the reported results.

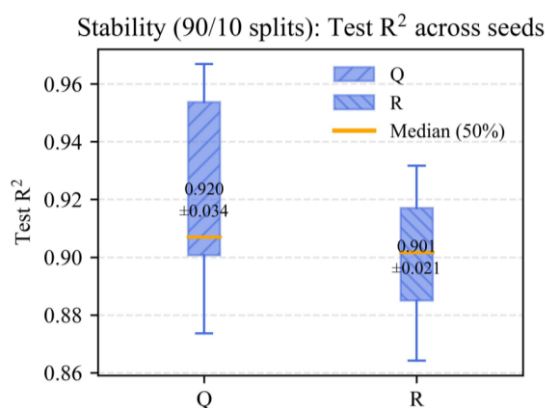


Fig. T3. Model evaluation results from repeating the train-test split with different random seeds 10 times

2. Subgroup evaluation across different target heavy metals. To further assess model robustness and potential heterogeneity in predictive performance, we conducted subgroup evaluations based on different target heavy metals in the final test set. Specifically, using the same trained model and fixed test set as in the main analysis, the test samples were partitioned into four subsets according to the adsorption target: Cr(VI), Pb(II), Cd(II), and As(V). For each subset, performance metrics including R^2 , MAE, and RMSE were computed independently. This constitutes a single-model, multi-subset evaluation framework (subgroup evaluation).

The results show that predictive performance varies across different targets. For Q prediction, the highest R^2 is observed for As(V) ($R^2 = 0.99$), while the lowest is for Cd(II) ($R^2 = 0.84$). For R prediction, the best performance is also achieved on As(V) ($R^2 = 0.94$), whereas Pb(II) yields the lowest performance ($R^2 = 0.87$). These results are visualized using eight scatter plots (Fig. T4 and Fig. T5).

This analysis addresses concerns regarding data heterogeneity and potential overestimation due to random splitting. While performance differences across subgroups are observed, the overall results remain consistent with the global test set performance ($R^2 = 0.95$ for Q and $R^2 = 0.90$ for R), indicating that the model maintains reliable predictive capability across different targets.

It should be noted that the performance on each individual subset is based on a fixed partition and may still be subject to sampling variability. Therefore, the overall test set evaluation, together with cross-validation results, should be considered as the primary indicators of model performance.

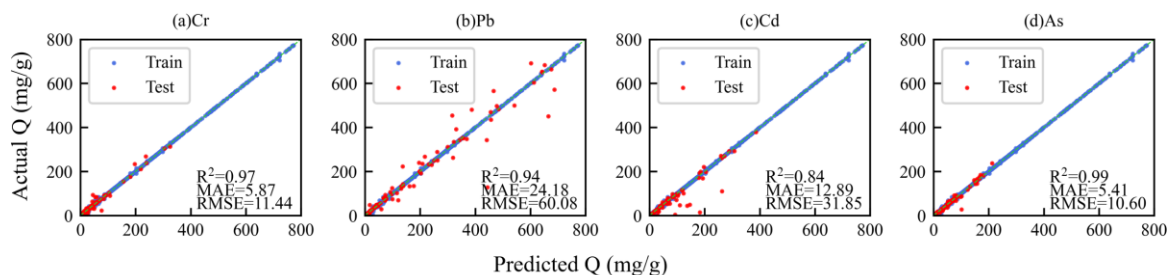


Fig. T4. Q prediction across heavy metal subgroups

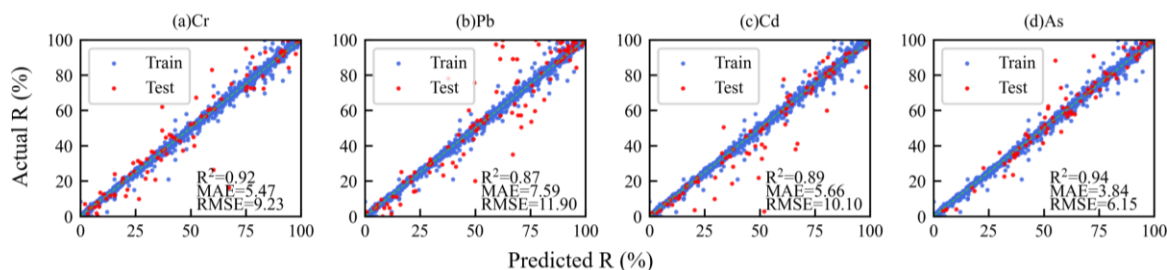


Fig. T5. R prediction across heavy metal subgroups

3. Stability analysis of SHAP-based feature importance. To evaluate the robustness of the

SHAP-based interpretability results with respect to data variability, we conducted a stability analysis under repeated data resampling, following a protocol consistent with the robustness analysis via repeated random splits. Specifically, for each target (Q and R), 10 independent repetitions were performed by randomly sampling 90% of the data. For each subset, the model was retrained and SHAP-based feature importance was computed.

The resulting feature importance rankings from each subset were then compared with those obtained from the full dataset using three quantitative metrics. As shown in Table T1, the feature importance rankings remain highly consistent across different subsamples. For Q, the Spearman correlation reaches 0.966 ± 0.027 with a Top-5 overlap of 1.000 ± 0.000 , while for R, the corresponding values are 0.973 ± 0.013 and 0.940 ± 0.092 . The L2 distances are also consistently low for both targets.

These results demonstrate that the SHAP-based interpretations are highly stable and not sensitive to specific data partitions. Despite the inherent dependence of SHAP values on both the model and the training data, the feature importance rankings exhibit minimal variation under repeated subsampling. This provides strong evidence that the interpretability conclusions reported in this study are robust and not driven by a particular data split.

Table T1. Stability of feature importance under repeated 90% data subsampling

	Subset fraction	Repeats	Spearman $\uparrow$	L2 $\downarrow$	Top-5 Overlap $\uparrow$
Q	0.9	10	0.966 ± 0.027	0.018 ± 0.004	1.000 ± 0.000
R	0.9	10	0.973 ± 0.013	0.019 ± 0.003	0.940 ± 0.092

Note. For each target (Q and R), 10 independent repetitions were performed by randomly sampling 90% of the data. Stability was quantified using three metrics: (i) Spearman’s rank correlation coefficient ($\uparrow$, higher = more stable), (ii) L2 distance between normalized importance vectors ($\downarrow$, lower = more stable), and (iii) Top-5 feature overlap ratio ($\uparrow$, higher = more stable). Values are reported as mean $\pm$ standard deviation across the 10 repeats.

Text S3. Performance and stability trade-off comparison of different imputation

strategies

Although the original data preprocessing employed a domain knowledge-based method for handling missing values, we also conducted additional comparable sensitivity analyses using other fully reproducible imputation strategies on the existing raw dataset. Taking the prediction of Q as an example, samples with missing Q values were directly excluded; using Tim as an auxiliary variable, samples with missing Tim values were also excluded (resulting in 3,587 remaining samples). Then, without any sample filtering process, comparative experiments of various imputation methods (mean, random forest, hybrid imputation) were conducted directly. Fixed imputation parameters were used, and the experiments were repeated ten times with random seeds.

The results showed that although minor performance differences were observed, our hybrid imputation method maintained high performance and stability. The importance patterns of key features remained consistent, indicating that the research findings do not depend on specific imputation schemes.

As shown in Fig. T6, we observe that mean imputation achieves the highest average test performance ($R^2 = 0.902$), followed by the hybrid strategy ($R^2 = 0.894$) and random forest imputation ($R^2 = 0.881$). However, mean imputation also exhibits the largest variability across random splits ($\text{std} = 0.044$), indicating higher sensitivity to data partitioning.

In contrast, the hybrid imputation strategy achieves slightly lower mean performance but demonstrates improved stability ($\text{std} = 0.034$), suggesting a more robust representation of the underlying data structure. This reflects a better balance between predictive accuracy and robustness.

Random forest imputation shows inferior performance, likely due to error propagation in estimating missing values under heterogeneous data conditions.

As shown in Table. T2, Our hybrid imputation method has an 90% overlap rate with the top-10 features of other methods, indicating a high consistency in the identified key factors. In addition, the Spearman rank correlation between feature importance rankings remains high (e.g., >0.85), further confirming the stability of feature relevance.

Overall, these results indicate that while simple imputation methods may yield higher predictive scores, the proposed hybrid strategy provides a more stable and physically consistent basis for downstream analysis.

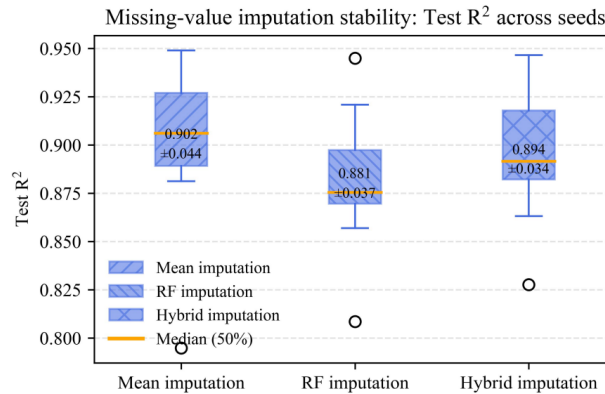


Fig. T6. Performance and stability trade-off comparison of different imputation strategies. Mean imputation achieves the highest predictive performance but exhibits the greatest variability across random splits, indicating lower stability. The hybrid imputation strategy shows slightly lower performance but improved stability, reflecting a better balance between accuracy and robustness. In contrast, random forest imputation performs worse in both predictive accuracy and stability.

Table T2. Stability of feature importance under different imputation strategies

Comparison object	Spearman $\uparrow$	Top-10 Overlap $\uparrow$
RF imputation vs Mean imputation	0.736	0.9
Hybrid imputation (Ours) vs Mean imputation	0.851	0.9
Hybrid imputation (Ours) vs RF imputation	0.877	0.9

Note. Spearman's rank correlation coefficient ($\uparrow$, higher = more stable), Top-10 feature overlap ratio ($\uparrow$, higher = more stable).

Text S4. Mechanistic consistency analysis with representative literature (e.g., CaFu and NU-1000)

To further examine whether the identified feature patterns reflect physically meaningful relationships, we compare the model-derived insights with representative literature cases.

As illustrated in Fig. 8, the sample exhibiting the highest Cd(II) adsorption capacity from water by CaFu MOF is visually explained (predicted value of 689.54 mg g⁻¹ and actual value of 781.2 mg g⁻¹). The AD has the highest SHAP value. Importantly, the three features related to the MOF are the key contributing features identified by the model that are associated with its favorable predicted performance. Thanks to the combined effect of Ca and FA, their SHAP contribution values rank fourth and second, respectively. The original study indicates that “Cd(II) ions interacted with MOFs via Cd(II) OeH non-covalent interactions. Cd(II) ions have been observed to interact with MOFs at various sites, including carboxylate anions coordinated with the Ca(II) centre. This interaction is facilitated by strong anion-cation interactions, which enhance the coordination of Cd(II)” . The PV is also an essential factor in increasing the adsorption capacity, ranking third. The partial dependence plot of feature PV by the MQ model (Fig. S4B) indicates a positive statistical association with the adsorption capacity, which increases with a rise in pore volume.

For Cr(VI) adsorption, the model consistently identifies Zr as a high-performing metal type (MT) across relevant scenarios. This observation aligns well with reported studies on Zr-based MOFs such as NU-1000, where superior adsorption performance is attributed to a high density of accessible Zr–OH active sites, which act as strong binding centers for dichromate ions. In addition, the large pore channels of NU-1000 facilitate rapid diffusion and accessibility to these active sites, while favorable electrostatic interactions under acidic conditions further enhance adsorption efficiency. Importantly, these experimentally supported mechanisms (active site density, pore accessibility, and electrostatic interactions) correspond closely to the key feature contributions identified by the model. We emphasize that such comparisons do not constitute independent validation, as these studies are part of the collected dataset. Instead, they serve as consistency evidence indicating that the model is able to recover known high-performance patterns and mechanistic trends from heterogeneous literature data.

Table S1. Abbreviation of organic ligands

Num	Abbreviation	Interpretation
1	Hmim	2-Methylimidazole
2	H ₂ BDC	terephthalic acid
3	NH ₂ -BDC	2-aminoterephthalic acid
4	H ₃ BTC	Trimesic acid
5	TATAB	4,4',4''-(1,3,5-triazine-2,4,6-triyltriimino) tris-benzoic acid (H ₃ TATAB))
6	DOBDC	2,5-Dihydroxyterephthalic acid(DOBDC)
7	3A5MT	3-amino-5-mercapto-1,2,4-triazole
8	BenA	benzoic acid
9	H ₂ BPDC	4,4' - Biphenyldicarboxylic acid
10	BPYDC	2,2' - bipyridine - 5,5' - dicarboxylic acid
11	MA	Mercaptosuccinic acid
12	H ₃ BTBA	4,4',4''-(1Hbenzo[d]imidazole-2,4,7-triyl)tribenzoic acid
13	35DTZ	3,5-Diphenyl-1,2,4-triazole
14	PTA	1, 4-dicarboxybenzene(PTA)
15	FA	fumaric acid,FA
16	Benz	benzimidazole
17	TipaL	Tipa&L-proline
18	H ₃ TATB	2,4,6-tris(4-carboxyphenyl)-1,3,5-triazine
19	5AmiI	5-amino-isophthalic acid
20	TCPBDA	N,N,N',N'-tetrakis(4-carboxyphenyl)biphenyl-4,4'-diamine(H ₄ TCPBDA)
21	DFSA	2,6-amidopyrimidine-2-hydroxy5-benzoic acid
22	BCA	1,2,4,5 Benzenetetra_x005f carboxylic Acid
23	H ₃ LBip	H ₃ L=4-(N,N'-bis(4-carboxybenzyl)amino) benzenesulfonic acid, 4,4'-bipy = 4,4'-bipyridine
24	TIB	1,3,5-tris(1-imidazolyl)benzene(tib)
25	g-CD	g-cyclodextrin (g-CD)
26	SUCA	succinate acid

Table S2. Abbreviation of modification method

Num	Abbreviation	Interpretation
1	AcLI	Acetylacetone loaded iron
2	CAC	Cellulose aerogel composite
3	GO	Modification with graphene oxide (GO)
4	PEI	Polyethylenimine (PEI) modified
5	MASC	Mineralizing agent sodium carbonate (Na_2CO_3)
6	TMAOH	Tetramethyl ammonium hydroxide (TMAOH)
7	HF	Hydrofluoric acid
8	Car	Carbonization
9	Cal	Calcination
10	AmiF	Amino-functionalization
11	BisM	Modification with Bisquinoline Mannich
12	Feo	Magnetic Fe_3O_4 microparticles
13	CMC	Carboxymethylcellulose-coated,CMC
14	EDA	Ethylenediamine
15	HieP	Hierarchically porous
16	AcOH	AcOH modified
17	HCL	HCL modified
18	FeCo	Fe/Co molar ratio of 2/1
19	Cnon	Carbon nano-onion embedded
20	SHG	Sulfhydryl group functionalized
21	SOA	Sulfonic acid group functionalized
22	MeOH	MeOH,incorporation of pendent sulphur-containing groups within the pores of the material
23	INO	Isonicotinic acid N-oxide (INO)
24	IL	Acidic chloroaluminate
25	MP	Magnetic polydopamine
26	HA	Alginate acid
27	MCSDE	MOF was grown on the surface of magnetic chitosan diatomite(M-CS-DE) by insitu growth method
28	MPC	Magnetic porous g- Fe_2O_3 /C@HKUST-1 composites
29	InFe	Incorporating Fe into the material
30	4TH	4-Thioureidobenzoicacid (Th) is rich in thionic groups
31	ITB	Coordination bonds of unsaturated Ti metal centres and the amine groups in 2-imino-4-thiobiuret (ITB)
32	DHAQ	2-Aminoterephthalic acid,1,8-dihydroxyanthraquinone (DHAQ)
33	CuFe	Fe doped Cu:Fe mole ratios (95:5)
34	Mela	Melamine modification
35	MITG	Immobilizing tripeptide glutathione on the surface of a magnetic metal organic framework
36	EDTA	Ethylenediaminetetraacetic acid (EDTA) functionalized
37	Polya	Decorated polyaniline

Num	Abbreviation	Interpretation
38	FA	Fumaric acid,FA
39	DCAA	Defect Control(monocarboxylic acid modulator[acetic acid(AA)])
40	DCTFA	Defect Control(monocarboxylic acid modulator[trifluoroacetic acid (TFA)])
41	LAM	(LAM)lanthanum-modified
42	CTAB	trimethylammonium bromide (CTAB) as a structure-directing agent.
43	AceM	Acetate modulated
44	EtOH	EtOH as solvent mixed with agate mortar
45	MPTS	Thiol-functionalized,3.2 g TEOS and 0.9 g (3-mercaptopropyl)-trimethoxysilane(MPTS)
46	GMA	Glycidyl methacrylate
47	ASA	Aminosulfonic acid functionalized
48	DEM	Dimethyl Ethylenediamine modification
0	UnMod	Unmodified MOFs

Table S3. Summary statistics of quantitative variables using for predictive models

Variables	Mean	Std Dev	Upp Qu	Median	Low Qu	Min	Max	Std Er
SA (m ² g ⁻¹)	882.92	759.90	1196.00	877.00	216.70	1.18	3800.00	12.50
PD (nm)	4.70	7.50	3.32	1.99	1.14	0.06	40.00	0.12
PV (cm ³ g ⁻¹)	0.51	0.55	0.64	0.41	0.28	0.00	7.87	0.01
RPM (rpm)	236.32	108.49	239.00	239.00	180.00	30.00	800.00	1.78
SV (mL)	46.24	68.90	50.00	20.00	10.00	4.00	500.00	1.13
CI (mg L ⁻¹)	118.13	181.00	104.05	50.00	19.35	0.05	1500.00	2.98
AD (mg L ⁻¹)	1268.72	3928.52	1000.00	600.00	250.00	5.00	50000.00	64.61
Tim (min)	453.15	678.49	480.00	150.00	45.00	0.00	4320.00	11.16
pH	5.91	1.86	7.00	6.00	5.00	0.98	13.00	0.03
Tem (K)	299.47	5.50	298.00	298.00	298.00	273.00	338.00	0.09
Q (mg g ⁻¹)	91.66	151.77	98.02	44.82	15.00	0.00	1982.14	2.50
R (%)	56.38	30.88	85.34	57.87	30.58	0.00	100.00	0.51

*Std Dev: Standard Deviation; Std Er: Standard Error; Upp Qu:Upper Quartile; Low Qu:Lower Quartile.

Table S4. Performance evaluation scores for MQ and MR with the increase in the number of features

Output target	Test set	Number of features									
		5	6	7	8	9	10	11	12	13	14
Q	MAE	28.72	25.43	25.12	22.24	21.11	21.14	18.71	18.77	13.87	12.71
	RMSE	59.02	56.31	56.55	50.58	48.58	48.67	46.48	46.36	38.35	36.61
	R^2	0.861	0.874	0.873	0.898	0.906	0.906	0.914	0.914	0.944	0.947
R	MAE	18.36	16.21	14.49	12.76	12.64	12.64	9.59	9.31	6.52	5.77
	RMSE	23.71	21.93	20.13	17.96	17.82	17.89	14.66	14.19	11.16	9.75
	R^2	0.423	0.506	0.584	0.669	0.674	0.672	0.779	0.793	0.872	0.903

Table S5. The results of 5-fold cross-validation with parameter tuning on the training set

Best hyperparameters							
	n_estimators	colsample_bytree	gamma	learning_rate	max_depth	min_child_weight	subsample
MQ	200	1	0.5	0.1	15	1	0.9
MR	200	1	0.4	0.1	15	3	0.7
5-fold CV (R^2 _score)							
	1	2	3	4	5	mean	std
MQ	0.9379	0.9678	0.8828	0.9575	0.9657	0.9423	0.0316
MR	0.9069	0.8890	0.8985	0.8609	0.8992	0.8909	0.0160

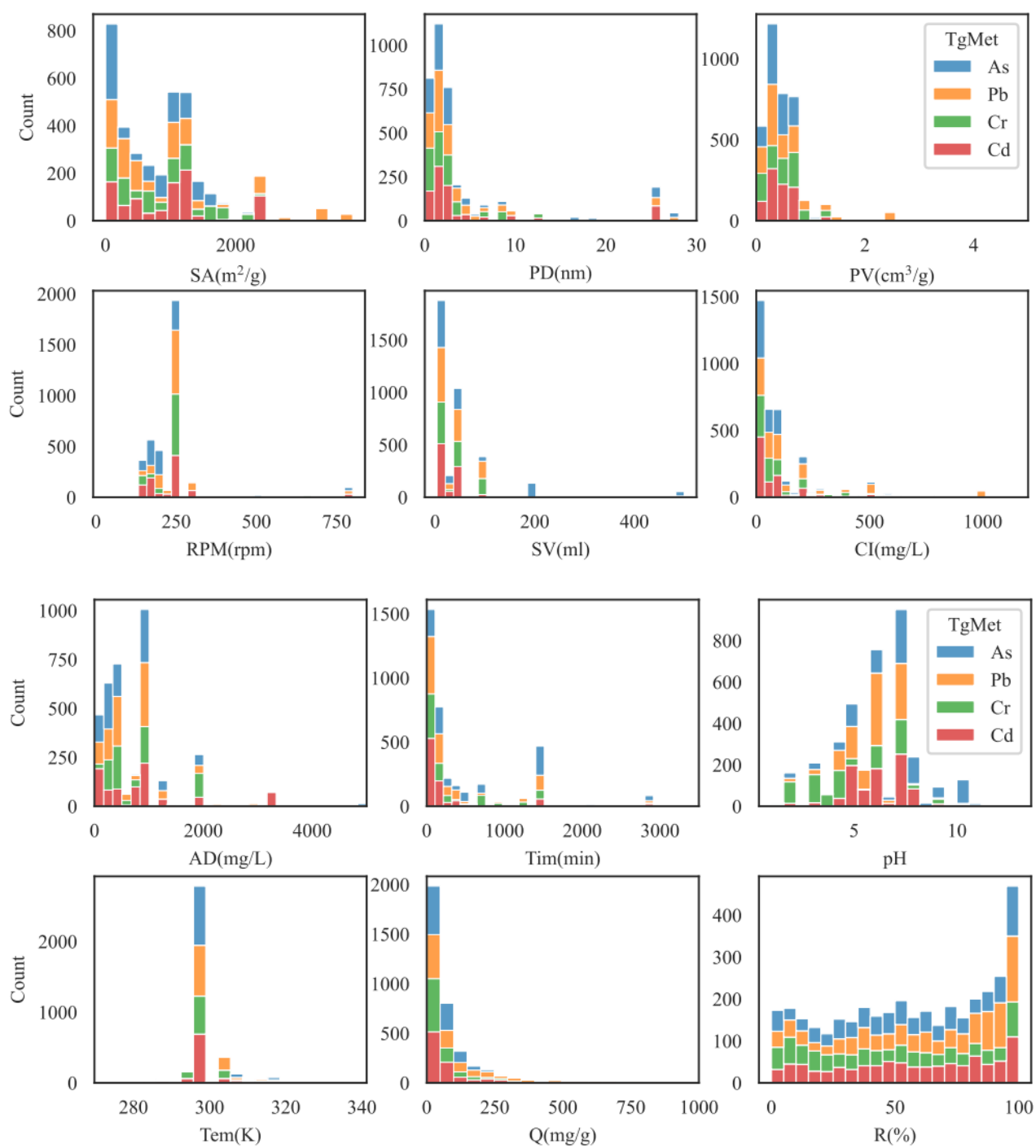


Fig. S1. Statistical distribution of quantitative variables

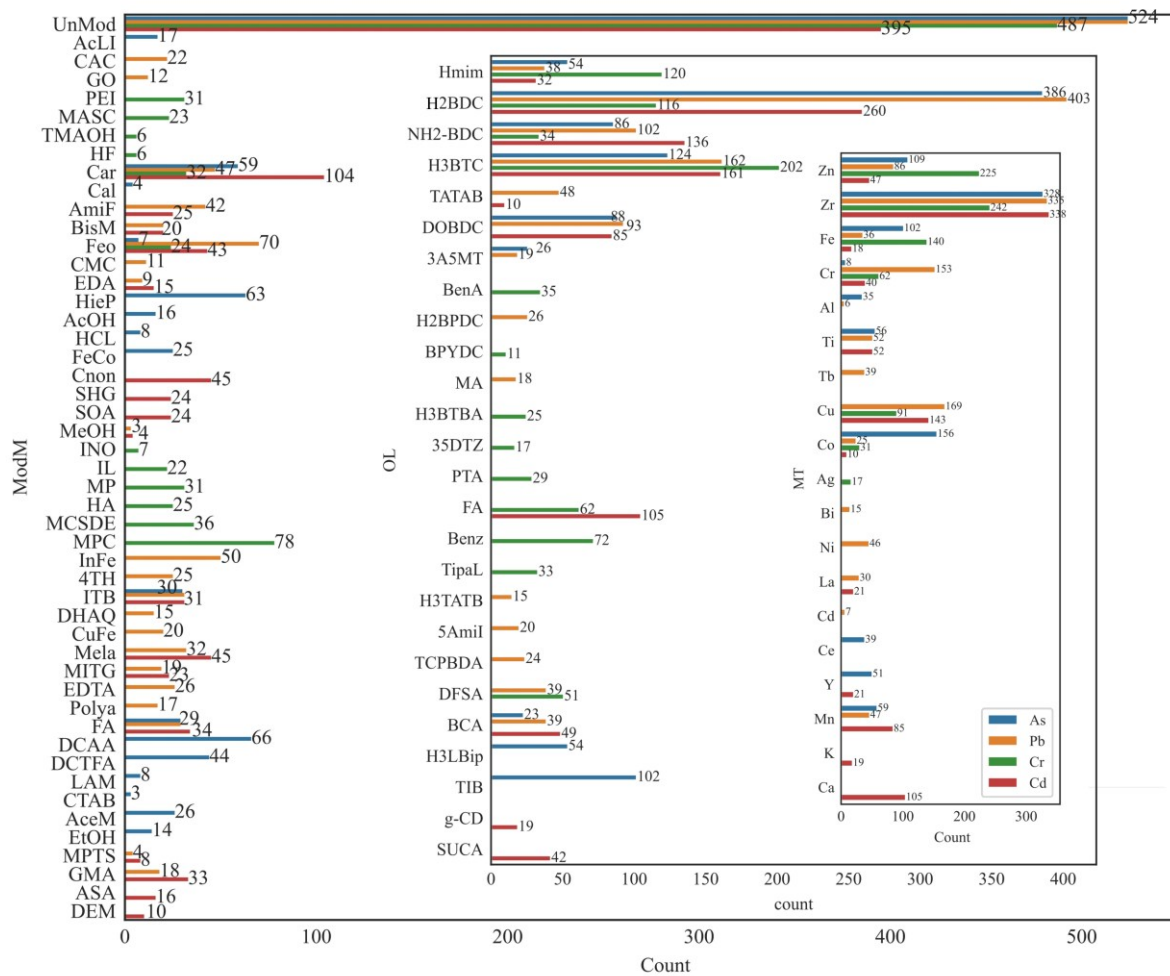


Fig. S2. Statistical distribution of qualitative variables

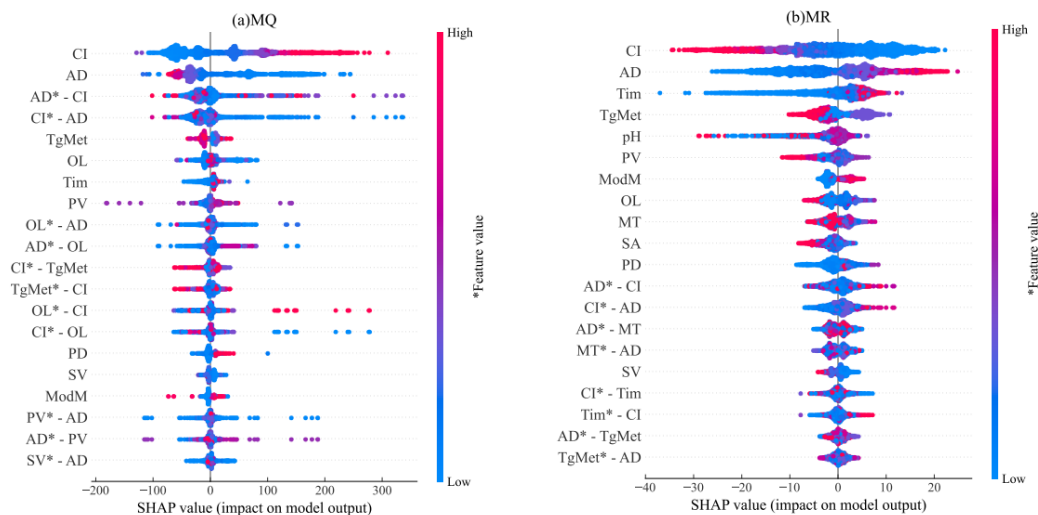
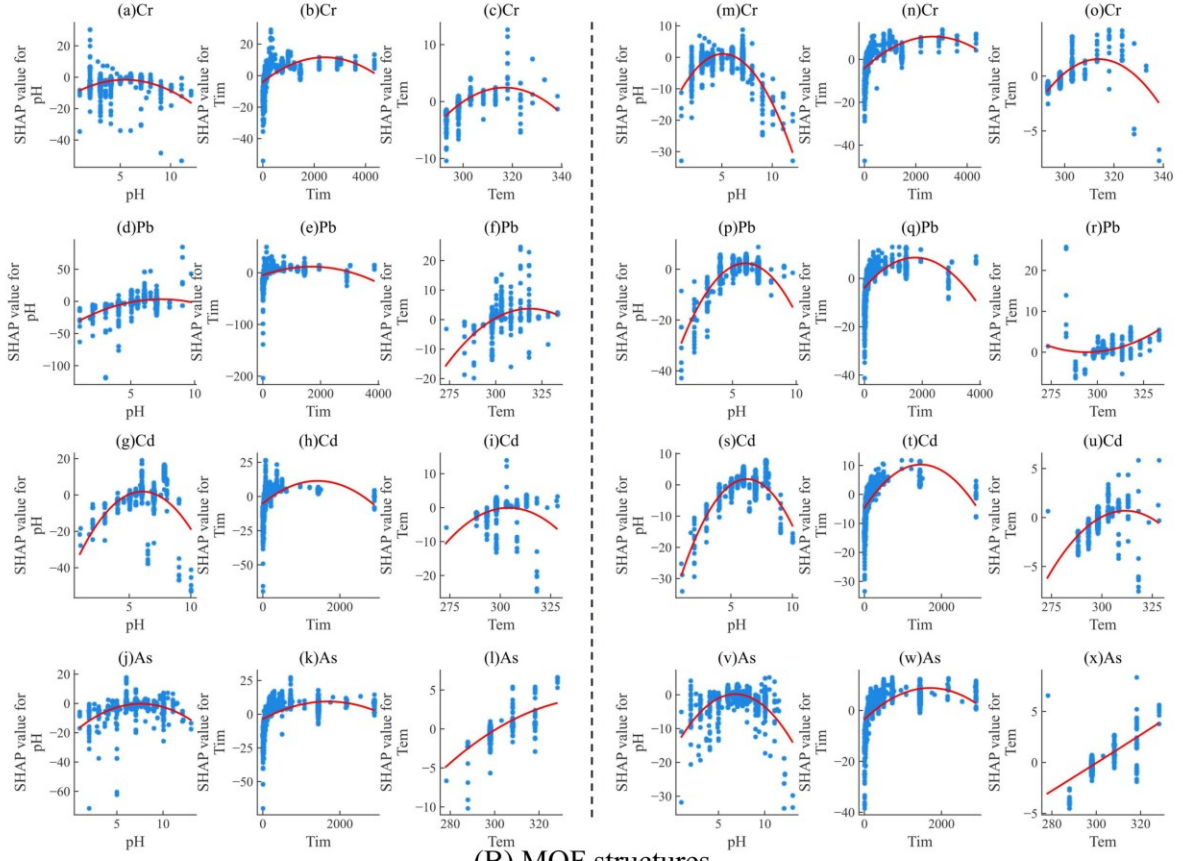


Fig. S3. SHAP based feature interaction importance. Two bee swarm plots are used to display the importance of feature interaction, where each point represents a sample from the data set. The position along the x-axis reflects the impact of the corresponding features on the model prediction for a specific sample. When multiple points overlap on the same x-axis position, they are stacked to demonstrate the density.

(A) Experimental conditions



(B) MOF structures

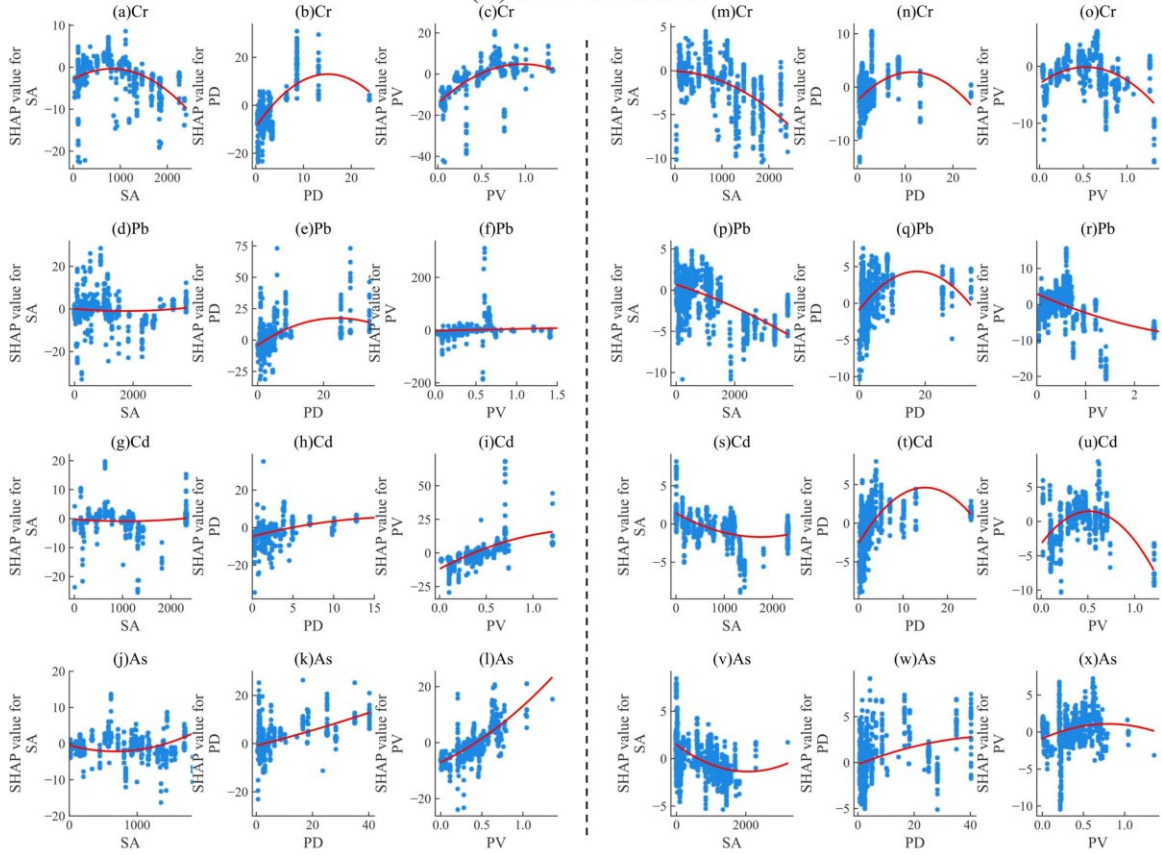


Fig. S4. Partial dependence based on SHAP. (A) Partial dependence of experimental conditions. (B) Partial dependence of MOF structures. (a-l) Partial dependence of MQ. (m-x) Partial dependence of MR. Each dot is a single prediction (row) from the dataset. The x-axis is the value of the feature. The y-axis is the SHAP value for that feature, which represents how much knowing that feature's value changes the output of the model for that sample's prediction. The red line is the trend reference line for the SHAP value.

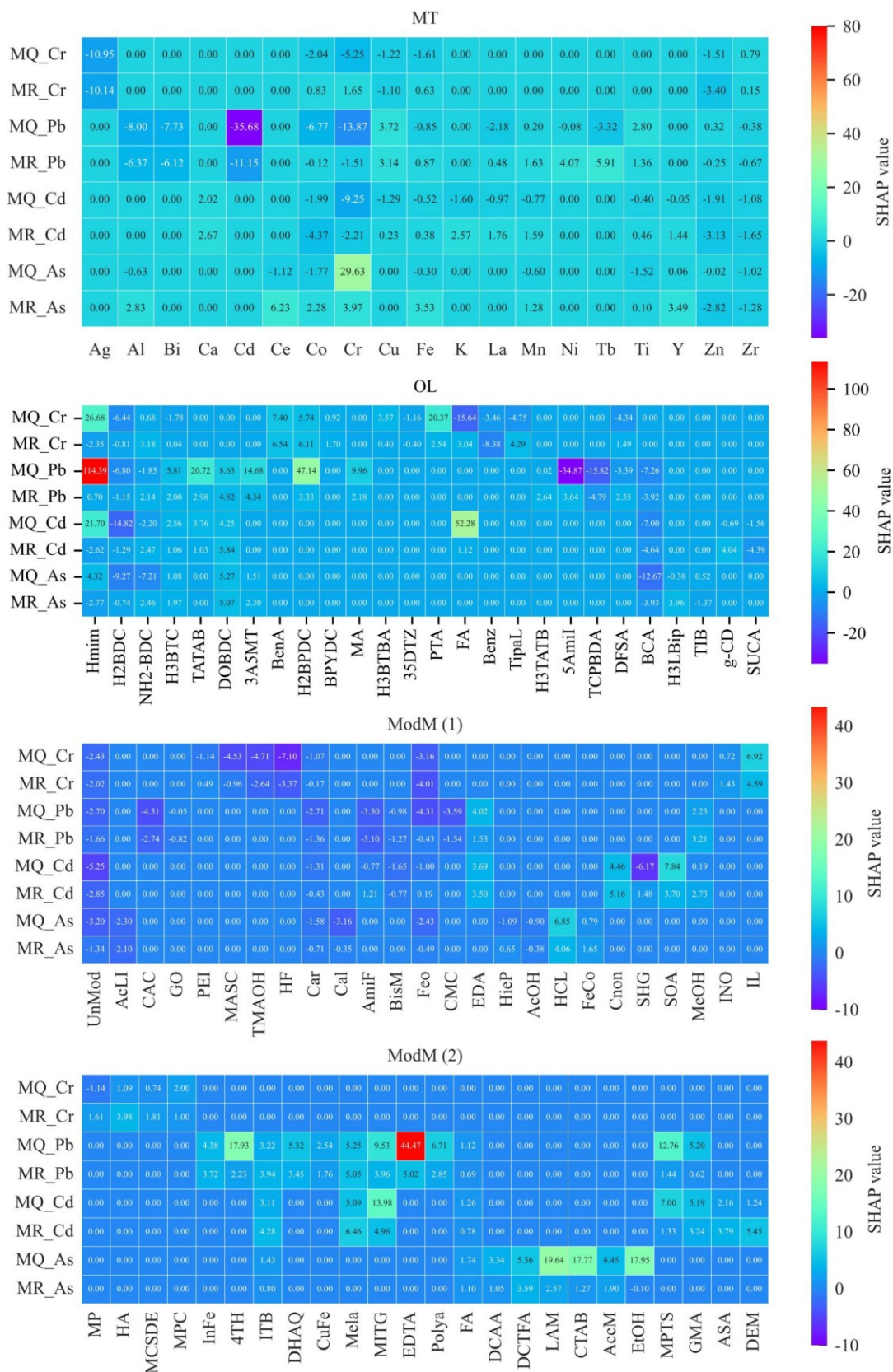


Fig. S5. Matrix of SHAP values for MT, OL & ModM. MQ/MR_Cr/Pb/Cd/As represents the

target heavy metals adsorbed by MOFs and the performance index Q or R. The SHAP value represents the average of all SHAP values for the corresponding feature value in a particular application scenario. If the SHAP value is 0.00 it means that the feature value does not exist in the particular application scenario.